

```
cvt 1680 1050 60
```

this gives you:

```
# 1680x1050 59.95 Hz (CVT 1.76MA) hsync: 65.29 kHz; pclk: 146.25 MHz
Modeline "1680x1050_60.00" 146.25 1680 1784 1960 2240 1050 1053 1059 1089 -
hsync +vsync
```

Now tell this to **xrandr**:

```
xrandr --newmode "1680x1050_60.00" 146.25 1680 1784 1960 2240 1050 1053 1059
1089 -hsync +vsync
```

Then you can now add it to the table of possible resolutions of an output of your choice:

```
xrandr --addmode VGA-0 1680x1050_60.00
```

The changes are lost after reboot, to set up the resolution persistently, create the file
~/.xprofile with the content:

```
#!/bin/sh
xrandr --newmode "1680x1050_60.00" 146.25 1680 1784 1960 2240 1050 1053 1059
1089 -hsync +vsync
xrandr --addmode VGA-0 1680x1050_60.00
```

To make it persistent, just add the resulting xrandr commands to your ~/.xprofile :

```
#!/bin/sh
xrandr --newmode "1920x1080_60.00" 172.80 1920 2040 2248 2576 1080 1081 1084
1118 -HSync +Vsync
xrandr --addmode VGA1 "1920x1080_60.00"
xrandr --output VGA1 --mode "1920x1080_60.00"
```

```
#!/bin/bash
```

```
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#
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```

```

# SOFTWARE.
#
# ---
# Q: WHY IS THERE A COPYRIGHT NOTICE ON THIS SCRIPT?
# A: It had to be added on (a very reasonable) user request.
# https://gist.github.com/chirag64/7853413#gistcomment-3736041

# First we need to get the modeline string for xrandr
# Luckily, the tool `gtf` will help you calculate it.
# e.g. `gtf <hRes> <vRes> <refreshRate>`:
gtf 1920 1080 60

# In this case, the horizontal resolution is 1920px the
# vertical resolution is 1080px & refresh-rate is 60Hz.
# IMPORTANT: BE SURE THE MONITOR SUPPORTS THE RESOLUTION

# Typically, it outputs a line starting with "Modeline"
# e.g. "1920x1080_60.00" 172.80 1920 2040 2248 2576 1080 1081 1084 1118 -
# HSync +Vsync
# Copy this entire string (except for the starting "Modeline")

# Now, use `xrandr` to add a new display mode. Pass the
# copied string as the parameter to the --newmode option:
xrandr --newmode "1920x1080_60.00" 172.80 1920 2040 2248 2576 1080 1081 1084
1118 -HSync +Vsync

# Well, the string within the quotes is the nick/alias
# of the display mode - you can as well pass something
# as "MyAwesomeHdResolution". But, why though!?!

# Then all you have to do is to add the new mode to the
# display you want to apply, like this:
xrandr --addmode VGA1 "1920x1080_60.00"

# VGA1 is the display name, it might differ for you.
# Run `xrandr` without any parameters to be sure.
# The last parameter is the mode-alias/name which
# you've set in the previous command (--newmode)

# It should add the new mode to the display & apply it.
# If it doesn't apply automatically, force it with:
xrandr --output VGA1 --mode "1920x1080_60.00"

# That's it... Enjoy the new awesome high-res display!

# NOTE to make the change persistent over reboots:
# - save the script file (with the necessary changes)
# - run it at startup (search the web for "How To")
# Thanks for the feedback!

```